

Parent User Stories

1. Authentication s Account Management

- As a parent, I can create an account using my email and password, and log in later, so I can protect my child's data and access it securely.
- As a parent, I can log in using Google so I can access the platform quickly and easily.
- As a parent, I can recover my password through email so I do not lose access to my account.

2. Child Profile Management

- As a parent, I can add a child by entering their name, age, and some initial notes so the AI can build a preliminary profile about their condition.
- As a parent, I can add more than one child under the same account so I can manage and monitor all of them.
- As a parent, I can edit my child's information at any time so the data remains accurate.

3. AI Behavioral Analysis s Personalized Planning

- As a parent, I can upload a short video of my child so the AI can analyze behaviors and extract useful behavioral insights.
- As a parent, I can answer a guided questionnaire about my child's behavior so the AI can build an accurate understanding of their condition.
- As a parent, I can view a personalized daily plan that includes scheduled activities and timings so I know exactly what to do with my child.

4. Daily Activity Tracking s Progress Monitoring

- As a parent, I can view today's activities organized by time on the homepage so I do not miss any activity.

- As a parent, I can view charts that show my child's progress weekly and monthly so I can monitor improvement over time.
- As a parent, I can export a comprehensive PDF report containing my child's profile and progress, highlighting strengths and areas for improvement, so I can share it with doctors or specialists for a quick and clear understanding of my child's condition.

5. Community's Parent Support

- As a parent, I can view posts from other parents and interact with them so I do not feel isolated.
- As a parent, I can share my experience or ask questions and receive responses from others so I can benefit from their experiences.
- As a parent, I can report inappropriate content so the environment remains safe and supportive.

6. AI Chatbot Assistant

- As a parent, I can open the chatbot and ask questions related to my child's behavior so I can better understand what is happening and how to respond appropriately.
- As a parent, the chatbot redirects me to a nearby specialist or center when needed if the question is beyond its scope, ensuring I do not receive inaccurate or inappropriate information.

7. Autism Centers Map's Discovery

- As a parent, I can open the map page and view all autism centers in Saudi Arabia, color-coded by type, helping me distinguish between public and private and charity centers.
- As a parent, I can click on any center on the map to view a card containing its name, contact number, working hours, location, and provided services.
- As a parent, I can search for a center by name or city to reach it directly without browsing the map.
- As a parent, I can filter centers by type (public, private, charity).
- As a parent, I can filter centers by city or region so distant centers that are difficult to access do not appear.
- As a parent, I can save any center to a favorites list for quick access later.

AI Functional Requirements

1. Video Analysis Engine

- As an AI, I analyze the uploaded video frame-by-frame and detect indicators such as eye contact, body movement, responses to stimuli, and repetitive behaviors so I can build an accurate behavioral profile.
- As an AI, I classify each observed behavior into specific categories (communication, social interaction, sensory behavior, motor skills) so the results remain organized and easy to follow.
- As an AI, I request the video to be re-uploaded if the lighting is poor or the child is outside the frame so I do not provide inaccurate analysis.
- As an AI, I generate a simplified text summary of the analysis in language understandable to parents without complex terms so the results can be applied directly.

2. Behavioral Insights from Questionnaires

- As an AI, I analyze questionnaire responses and extract daily behavioral patterns to support and improve understanding of the behaviors shown in the video while connecting observations that may not appear visually.
- As an AI, I give higher weight to behaviors repeatedly emphasized by the parent so the plan reflects the family's actual priorities.

3. Personalized Planning System

- As an AI, I select the most suitable activities from the activity library based on the child's level and weakest developmental areas so each session has a clear objective.
- As an AI, I schedule activities at appropriate times (morning, afternoon, evening) while considering rest periods, and avoid placing intensive activities immediately after meals so the plan remains realistic.
- As an AI, I define for each activity its goal, duration, required tools, and execution steps so the parent does not need external research.
- As an AI, I reassess the plan weekly based on progress data and automatically adjust ineffective activities to ensure the plan evolves continuously with the child.

- As an AI, I consider the parent's capabilities (available time and household resources) when building the plan so it becomes realistically applicable.
- As an AI, I detect signs of regression (such as increased withdrawal or stress), immediately adjust the plan, and notify the parent so unsuitable activities are not continued.
- As an AI, I identify activities where the child shows positive responses and increase them in future plans to strengthen the child's strengths.

4. Conversational AI Assistant

- As an AI, I distinguish between educational questions (which I answer directly) and medical or diagnostic questions (which I redirect to specialists) so I do not exceed my scope.

5. Reporting Insights System

- As an AI, I gather all child-related data (indicators, plans, progress, and chatbot notes) and organize them into a logical and chronological report so specialists can fully understand the child's development in one place.
- As an AI, I write a one-page narrative summary of the child's condition in clear language understandable to specialists (without AI-specific technical terminology) so the report is quick to read and easy to understand.
- As an AI, I include recommendations based only on collected data (not general opinions), while explaining the evidence or observations supporting each recommendation so they remain credible and reliable for specialists.

6. AI Safety, Privacy s Ethics

- As an AI, I never provide any medical or psychological diagnosis for the child under any circumstance, and instead only describe behaviors and observations objectively without diagnostic conclusions.
- As an AI, I always indicate the confidence level of every result or analysis, and warn when confidence is low so outputs are not treated as absolute facts.
- As an AI, I ignore any attempts to bypass instructions or requests outside the application's scope, such as medical prescriptions or restricted information.
- As an AI, I commit to protecting the child's data privacy and never share or transmit any external data except within a secure system, with encryption and clear consent.